# Echo Voyager/Orca XLUUV

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## Type

Extra-large unmanned underwater vehicle (XLUUV).

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## Key facts

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Echo Voyager is a fully autonomous extra-large UUV (XLUUV) designed to undertake a variety of missions such as intelligence, surveillance, and reconnaissance (ISR), in addition to applications in civil engineering and commercial exploration.

*Echo Voyager seen during its unveiling in March 2016. (Boeing)*

*1633911*

2024-02-12

**Development**

Development of the Echo Voyager began in 2013 with the objective of creating a XLUUV that could carry multiple modular payloads for a range of different mission profiles.

In 2017 the Echo Voyager underwent further development at Boeing facilities in Huntington Beach, California, and began initial sea trials off the coast of California in June the same year.

Boeing has also been contracted by the US Navy (USN) to design a new XLUUV designated Orca, which will be based on the Echo Voyager with some alterations to meet the service's requirements.

The USN stated that initial work will centre on payload integration and mine warfare missions. It is envisioned that such a large and flexible platform could contribute to mine countermeasures, anti-submarine warfare, anti-surface warfare, electronic warfare, and strike missions in the mid- to long-term. A USD43 million modification was added to the contract on 13 February 2019, covering the fabrication, test, and delivery of four Orcas.

The contract was further amended with an extra USD46.7 million awarded on 28 March 2019, adding an additional Orca to the order. This contract modification award completes the XLUUV competition and brings the total awarded amount for five XLUUVs and associated support elements to USD274.4 million.

On 17 September 2019, a USD10.8 million modification was added to the contract covering engineering services to further develop and study additional capabilities, payloads, operational uses, and deployment scenarios for the XLUUV programme.

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The Naval Undersea Warfare Division Keyport was selected on 4 August 2021 as the In-Service Engineering Agent (ISEA) for the Orca-class UUV, Large Displacement UUV, and Razorback dry deck shelter launch and recovery UUV, and will be the principal warfare centre responsible for the life-cycle sustainment of the aforementioned UUVs.

The XLUUV project has experienced some production delays. The Orca units, which were originally scheduled to be delivered in December 2020, are now planned to be finished in 2024. On 28 April 2022, the USN Naval Sea Systems Command (NAVSEA) revealed the first images of the lead Orca XLUUV Test Asset System being christened in Huntington Beach, California.

The Orca XLUUV commenced operational testing in March 2023, and the first unit was delivered to the US Navy in December 2023. Forward deployment is expected by 2026.

Orca XLUUV has been selected as the carrier vehicle for the NAVSEA Hammerhead encapsulated mine project.

*First Orca XLUUV prototype being launched by NAVSEA on 28 April 2022. (NAVSEA)*

*2010122*

2024-06-07

**Description**

The Echo Voyager can operate for months at a time in congested waters without physical human contact. It can also be launched and recovered without the support ships that normally assist UUVs. The vehicle can collect data while at sea and then surface to transmit the information.

**Design**

Echo Voyager is based on a cylindrical hullform somewhat similar to Boeing's Echo series AUVs such as Echo Ranger and Echo Seeker. It is designed to incorporate a modular payload section for multiple uses and can include payloads extending outside of its hull.

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The vehicle is designed to operate at sea for up to six months at a time. Its modular payload bay is fitted with commercial interfaces to accommodate various payloads and facilitate technology insertions, with electronics housed in the rear section.

Echo Voyager is equipped with a retractable mast that unfolds from the submersible and extends above the water to provide air for the generators to recharge the platform's batteries. Maintenance can be carried out at the pier, with subsystems that can be removed and swapped out with replacements while the vehicle remains in the water or on the dock.

**Propulsion**

Echo Voyager's powerplant comprises a hybrid diesel-electric system and a hybrid rechargeable power system which incorporates commercial off-the-shelf lithium-based batteries or silver zinc batteries. Additionally, Echo Voyager carries 3,785 litres (1,000 gallon) of diesel fuel that enables it to travel up to 12,038 km (6,500 n miles) on a single-fuel module. Endurance could be doubled by adding fuel tanks to the payload space. The XLUUV can operate for up to three days underwater with full propulsion and sensor use before it must surface for fresh air to recharge the batteries. When power in the batteries becomes low, the Echo Voyager surfaces and uses its diesel generator to recharge the batteries. It can travel approximately 150 n miles between recharges. The vehicle features an active buoyancy control system with the following features:

- Autonomous buoyancy control capability
- Forward and aft trim control
- Seafloor mooring capability

**Organic sensors and mission equipment**

Echo Voyager is equipped with a forward-looking sonar (FLS) for obstacle avoidance. Navigation is provided by a Kalman-filtered inertial navigation unit, a set of DVLs, depth sensors and a long baseline transponder. The vehicle's FLS system along with a DVL enable terrain following mission capability operations near the seafloor. A GPS aids in localisation when conducting operations at the surface.

**Payload**

The vehicle has a 10.3 m long and 2.6 m wide payload space that can support a payload volume of 56.63 $m^3$ and could be used for minelaying, among other missions such as surveillance and electronic warfare. The vehicle can also accommodate external payloads. Echo Voyager can utilise various types of sonar payload including a Raytheon PROSAS PS60-6000 synthetic-aperture sonar to enhance its ocean-bottom mapping capability.

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**<u>Variants</u>**

**Orca XLUUV**

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Orca is a modular, open-architecture XLUUV based on the Echo Voyager base model. The 15.5 m core vehicle will be used for navigation, control and communication, and a 10-meter payload can be configured to MCM, ASW or ISR capabilities.

The first Orca XLUUV prototype will be a Concept of Operations (CONOPS ) model used to reduce risk and gain valuable knowledge of the the operation, sustainment, and vehicle performance of XLUUVs. According to the USN's concept of operations, the Orca XLUUV is intended to transit an area of operation, loiter with the ability to periodically establish communications, deploy payloads, and transit home. Boeing has previously disclosed that the Orca is derived from its commercial Echo Voyager platform, which has completed more than 10,000 hours of at-sea testing since the type first embarked on system evaluations and testing in 2017.

The key differences between the Orca XLUUV and the Echo Voyager will be in the endurance, payloads, lift and building materials.

2024-06-07

**Specifications**

| | **Echo Voyager** | **Orca** |
|---|---|---|
| **Length:** | 15.50 m (without payload); 26 m (with payload) | 15.50 m (without payload); 26 m (with payload) |
| **Width:** | 2.60 m | 2.60 m |
| **Height:** | 2.60 m | 2.60 m |
| **Weight:** | 45,360 kg (in air) | 45,360 kg (in air) |
| **Operating depth:** | 3,000 m | 3,000 m |
| **Speed:** | 8 kt (3 kt optimal) | 8 kt (3 kt optimal) |
| **Endurance:** | >3 months | >3 months |
| **Range:** | 280 km (between recharges) at nominal speed | 280 km (between recharges) at nominal speed |
| **Communications:** | Inmarsat IV, Iridium, Wi-Fi, and freewave-enabled communications | Inmarsat IV, Iridium, Wi-Fi, and freewave-enabled communications(to be announced) |
| **Energy:** | 18 kW battery | 18 kW battery |

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**Status**

Echo Voyager – In production.Orca XLUUV – Prototype, in development/trials

2024-06-07

**Contractor**

Boeing Defense, Space & Security.

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